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MÜNCHEN

Software Engineering in Industrial Practice (SEIP)

Dr. Ralf S. Engelschall

ARCHITECTURE
FUNDAMENTALS

Architecture Stargate

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AF 01.1

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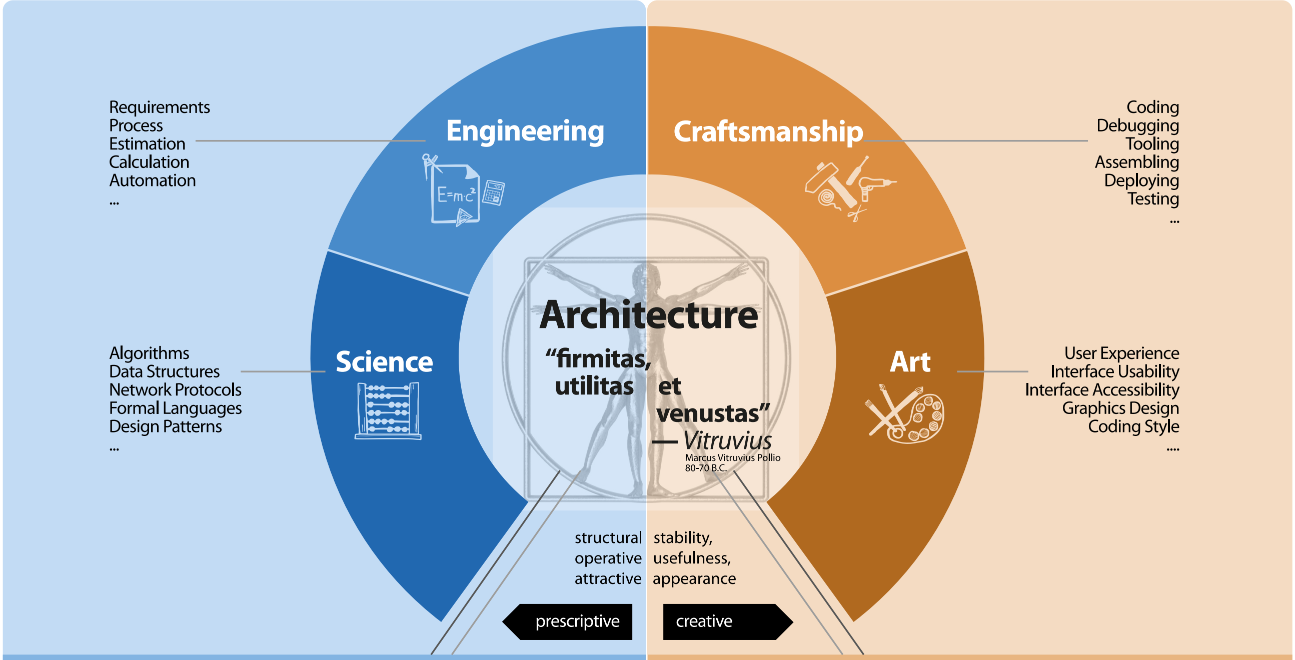

Requirements
Process
Estimation
Calculation
Automation
...

Engineering

Craftsmanship

Coding
Debugging
Tooling
Assembling
Deploying
Testing
...

Algorithms
Data Structures
Network Protocols
Formal Languages
Design Patterns
...

Science

Architecture

"firmitas, utilitas et venustas"
— *Vitruvius*
Marcus Vitruvius Pollio
80-70 B.C.

Art

User Experience
Interface Usability
Interface Accessibility
Graphics Design
Coding Style
....

structural stability,
operative usefulness,
attractive appearance

prescriptive

creative

Structural Scientific Definition of Architecture:
Architecture of a system is the set of fundamental **concepts** and **properties** of the system in its environment, embodied in its **elements**[, **interfaces**], **relationships**[and **behaviours**] and the principles of its **design** and **evolution**.
(based on industry standard *ISO/IEC 42010*)

Holistic Artistic Definition of Architecture:
Great architecture is achieved **harmony** and **accord** of all **parts**, where you no longer can add, modify or remove anything without impairing the **whole**.
(based on a quote by *Leon Battista Alberti* on beauty)

Architektur ist nicht einfach zu definieren. Man kann Architektur sowohl **strukturell wissenschaftlich** über messbare Elemente (elements), Schnittstellen (interfaces) und Beziehungen (relationships), als auch **ganzheitlich künstlerisch** über "die Harmonie und den Einklang aller Teile" definieren. Die "Wahrheit" liegt in der Praxis irgendwo dazwischen, denn die beiden Extrema spannen einen breiten Raum auf, in dem alle Lösungen in der Praxis liegen.

Auf der strukturell wissenschaftlichen Seite definiert sich Architektur über die Aspekte **Science** (insbesondere Computer Science) und **Engineering** (insbesondere Software Engineering). Auf der ganzheitlich künstlerischen Seite definiert sich Architektur über die Aspekte **Craftsmanship** (Handwerk, insbesondere Programmieren) und **Art** (Kunst, insbesondere User Experience).

Fragen

- Wie kann man **Architektur** definieren?
- Über welche vier **Aspekte** kann man **Architektur** aufspannen?

Adequacy and Beauty

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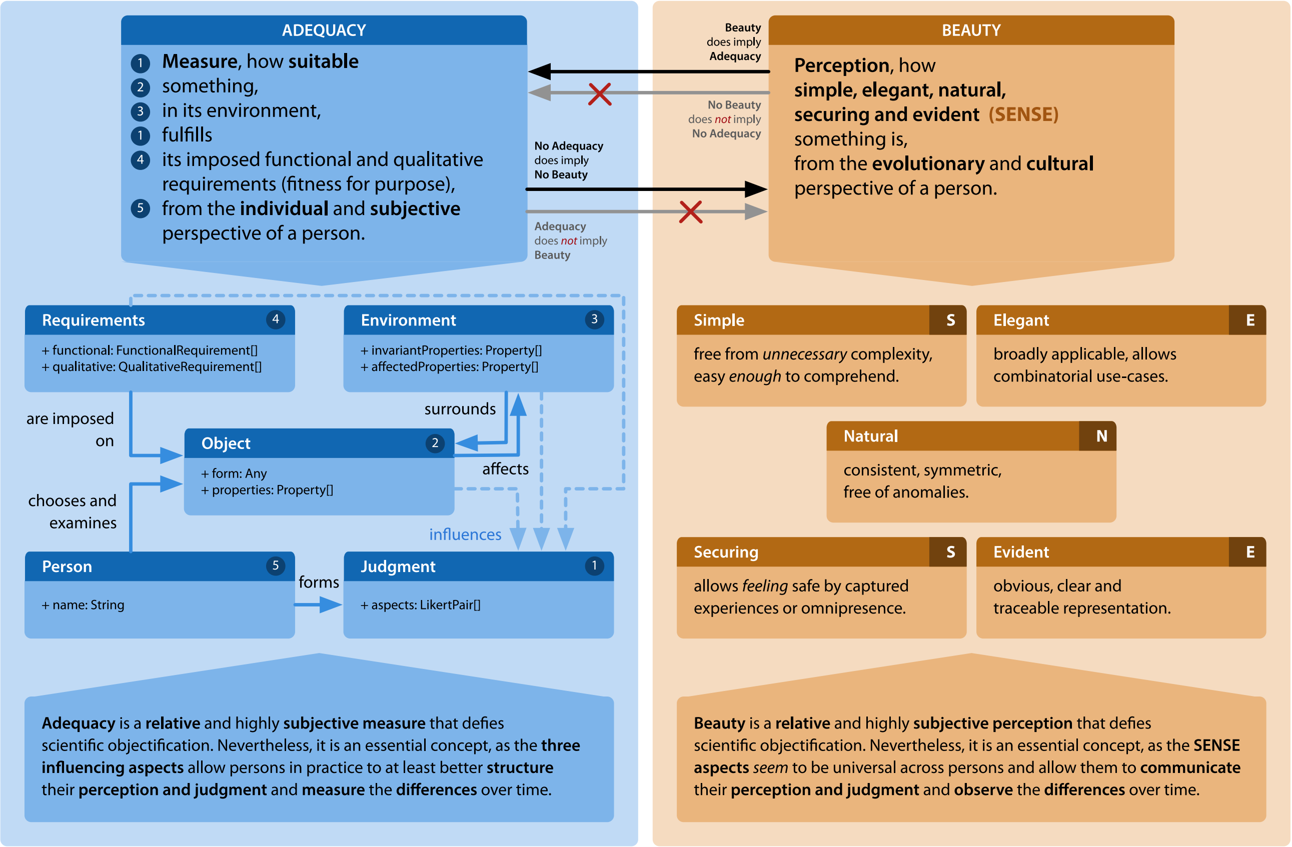

ADEQUACY

① **Measure**, how **suitable**
② something,
③ in its environment,
① fulfills
④ its imposed functional and qualitative requirements (fitness for purpose),
⑤ from the **individual** and **subjective** perspective of a person.

Beauty does imply **Adequacy**

No Beauty does *not* imply **No Adequacy**

No Adequacy does imply **No Beauty**

Adequacy does *not* imply **Beauty**

BEAUTY

Perception, how
simple, elegant, natural,
securing and evident (SENSE)
something is,
from the **evolutionary** and **cultural** perspective of a person.

Requirements ④
+ functional: FunctionalRequirement[]
+ qualitative: QualitativeRequirement[]

Environment ③
+ invariantProperties: Property[]
+ affectedProperties: Property[]

Object ②
+ form: Any
+ properties: Property[]

Person ⑤
+ name: String

Judgment ①
+ aspects: LikertPair[]

Simple S
free from *unnecessary* complexity, easy *enough* to comprehend.

Elegant E
broadly applicable, allows combinatorial use-cases.

Natural N
consistent, symmetric, free of anomalies.

Securing S
allows *feeling* safe by captured experiences or omnipresence.

Evident E
obvious, clear and traceable representation.

Adequacy is a **relative** and highly **subjective measure** that defies scientific objectification. Nevertheless, it is an essential concept, as the **three influencing aspects** allow persons in practice to at least better **structure** their **perception and judgment** and **measure** the **differences** over time.

Beauty is a **relative** and highly **subjective perception** that defies scientific objectification. Nevertheless, it is an essential concept, as the **SENSE aspects** *seem* to be universal across persons and allow them to **communicate** their **perception and judgment** and **observe** the **differences** over time.

Angemessenheit ist definiert als das Maß, wie geeignet etwas, in seiner Umgebung, die gestellten funktionalen und qualitativen Anforderungen erfüllt (Eignung für den Einsatzzweck), aus der individuellen Perspektive einer Person.

Angemessenheit ist ein relatives und höchst subjektives Maß, das sich einer wissenschaftlichen Objektivierung entzieht. Dennoch ist Angemessenheit ein wichtiges Konzept, da die drei beeinflussenden Aspekte (Anforderungen, Umgebung, Objekt) es den Personen in der Praxis ermöglichen, ihre Wahrnehmung und ihr Urteil zumindest besser zu strukturieren und die Unterschiede über die Zeit zu messen.

Schönheit ist definiert als die Wahrnehmung, wie einfach, elegant, natürlich, sicher und offensichtlich (SENSE) etwas ist, aus der evolutionären Perspektive eines Menschen.

Schönheit ist eine absolute und höchst subjektive Wahrnehmung, die sich einer wissenschaftlichen Objektivierung entzieht. Dennoch ist Schönheit ein wichtiges Konzept, da die SENSE-Aspekte universell über Personen hinweg zu sein scheinen und es ihnen erlauben ihre Wahrnehmung und ihr Urteil mitzuteilen und die Unterschiede über die Zeit zu messen.

Fragen

- Kann man Angemessenheit oder Schönheit allgemein messen?
- Kann man Angemessenheit oder Schönheit im Kontext einer einzelnen Person messen?

ARCHITECTURE FUNDAMENTALS

King Discipline Architecture

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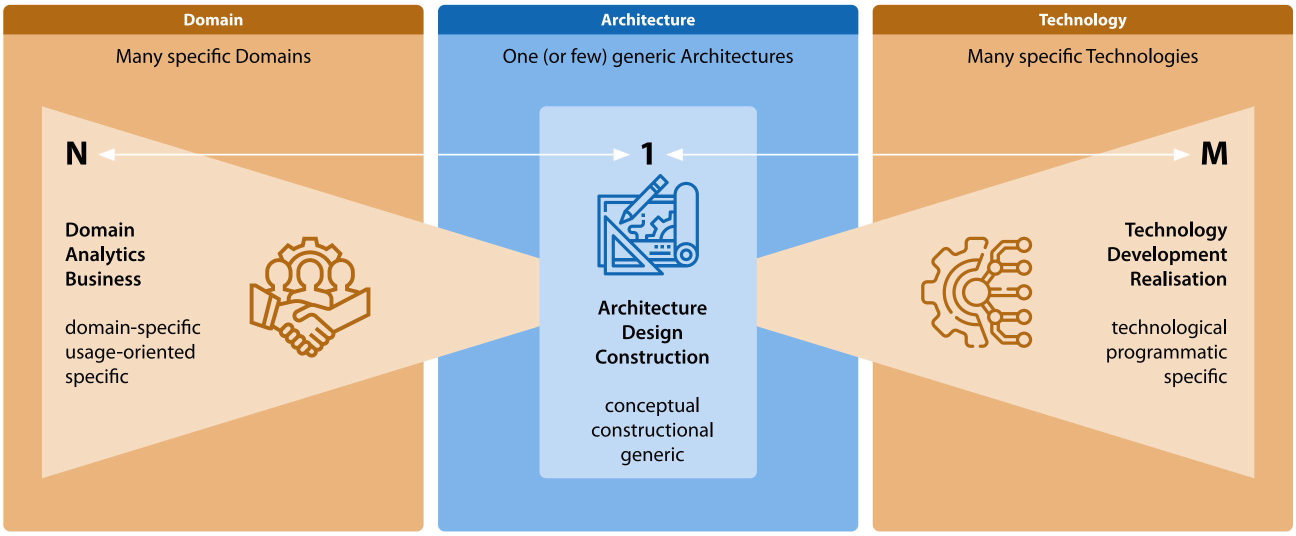


Domain

Many specific Domains

N

Domain
Analytics
Business

domain-specific
usage-oriented
specific

Architecture

One (or few) generic Architectures

1

Architecture
Design
Construction

conceptual
constructional
generic

Technology

Many specific Technologies

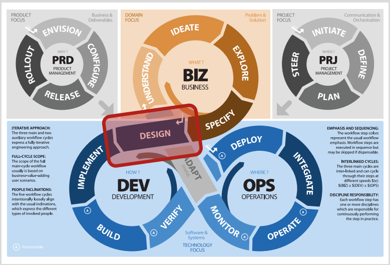
M

Technology
Development
Realisation

technological
programmatic
specific

Software Engineering Disciplines

Software Engineering Workflow

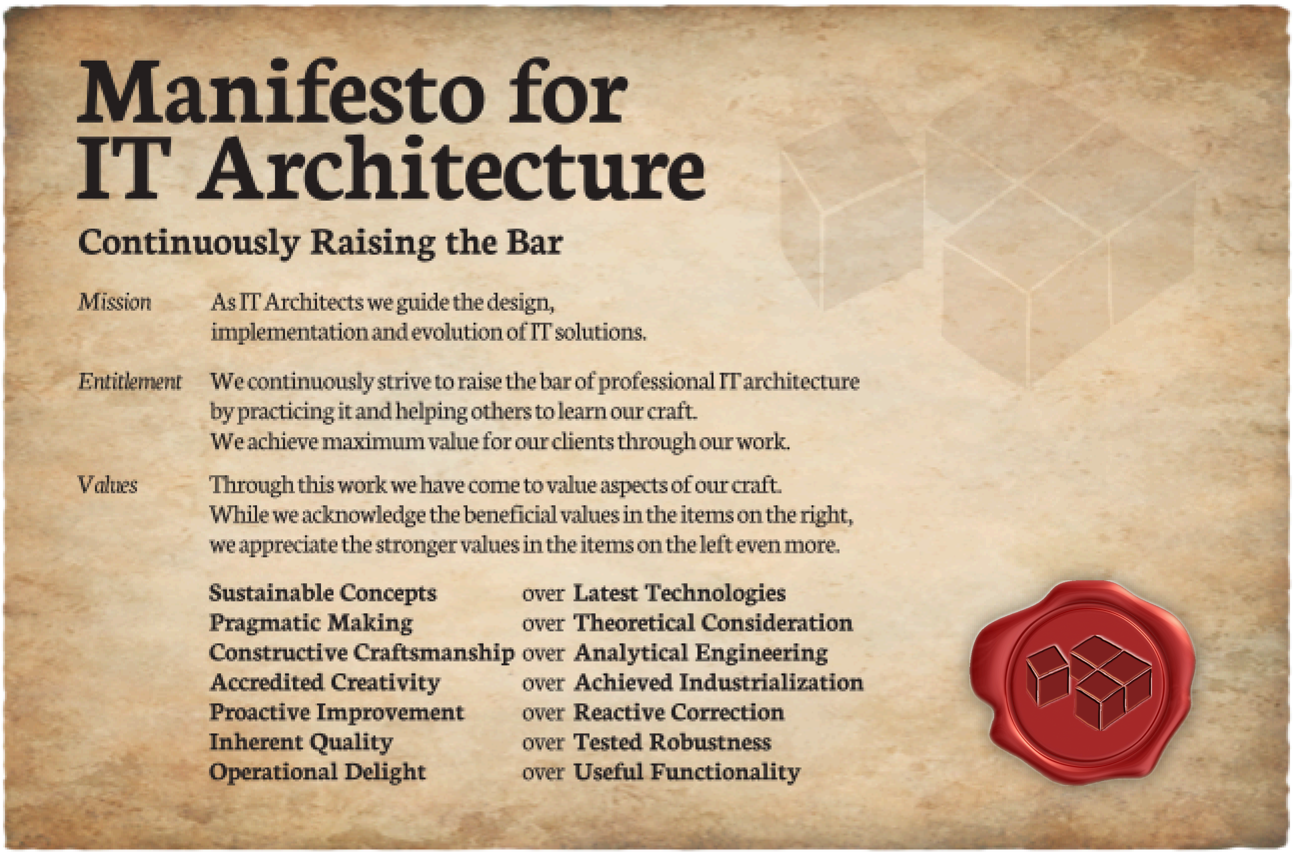


(Software) **Architecture** gilt als die “Königsdisziplin” im **Software Engineering**, da es das zentrale, allgemeine und konzeptuelle Bindeglied zwischen den vielen, potentiellen, spezifischen, realisierten **Fachlichkeiten** und den vielen, potentiellen, spezifischen, realisierenden **Technologien** ist. Die architekturelle Konstruktion einer Anwendung findet im logischen Schritt “Design” innerhalb des BizDevOps-Workflows im Software Engineering statt.

Fragen

- ? Wieso gilt **Architektur** als die “Königsdisziplin” im Software Engineering?

Architecture Manifesto



Manifesto for IT Architecture

Continuously Raising the Bar

Mission As IT Architects we guide the design, implementation and evolution of IT solutions.

Entitlement We continuously strive to raise the bar of professional IT architecture by practicing it and helping others to learn our craft. We achieve maximum value for our clients through our work.

Values Through this work we have come to value aspects of our craft. While we acknowledge the beneficial values in the items on the right, we appreciate the stronger values in the items on the left even more.

Sustainable Concepts	over	**Latest Technologies**
Pragmatic Making	over	**Theoretical Consideration**
Constructive Craftsmanship	over	**Analytical Engineering**
Accredited Creativity	over	**Achieved Industrialization**
Proactive Improvement	over	**Reactive Correction**
Inherent Quality	over	**Tested Robustness**
Operational Delight	over	**Useful Functionality**

Das **Manifesto for IT Architecture** ist eine Grundsatzerklärung für IT-Architektur. Sie besagt zuallererst, daß kontinuierlich die Meßlatte zu erhöhen ist (“Continuously Raising the Bar”), da nach gerade Mal 50 Jahren Software Engineering und Software Architecture wir zwar bereits etliche Best Practices kennen, aber die Disziplin sich sicherlich noch sehr lange weiterentwickeln muss.

Der Auftrag (Mission) an IT-Architekten ist, die Konstruktion, die Implementierung und die Weiterentwicklung/Wartung von IT-Lösungen zu leiten. Der Anspruch (Entitlement) ist es, dabei die Messlatte kontinuierlich zu erhöhen und anderen zu helfen, das “Handwerk” zu erlernen. Selbstredend ist die Tatsache, daß durch die Arbeit von Architekten maximaler Mehrwert für die Kunden erzielt wird.

Die Basis-Werte (Values), welche in diesem Handwerk eine zentrale Rolle spielen und sehr geschätzt werden, sind: **Latest Technologies**, **Theoretical Consideration**, **Analytical Engineering**, **Archieved Industrialization**, **Reactive Correction**, **Tested Robustness** und **Useful Functionality**.

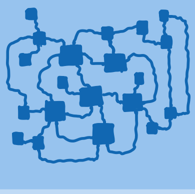
Darüber hinaus gibt es aber zusätzliche Werte, welche ebenfalls eine zentrale Rolle spielen und noch sehr viel mehr geschätzt werden: **Sustainable Concepts** (der Inhalt von **Architecture Fundamentals**!), **Pragmatic Making**, **Constructive Craftsmanship**, **Accredited Creativity**, **Proactive Improvement**, **Inherent Quality** und **Operational Delight**.

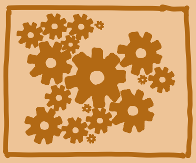
Fragen

? (keine)

Complex vs. Complicated

	complex	**complicated**
FOCUS	refers to the **extrinsic** and **higher**- or **macro**-level difficulty of a system,	refers to the **intrinsic** and **lower**- or **micro**-level difficulty of a system,
RATIONALE	because the system involves many different and **connected** parts	because the system involves many different and **difficult** aspects
CHALLENGE	which take time to **comprehend** and **master** in **total**,	which take time to **understand** and **learn** in **detail**,
INSIGHT	and which nevertheless are **easy** to explain.	and which usually are **hard** to explain.





	complex	complicated
NOTICE	**Simple** (non-complicated) systems can be **complex**.	**Clear** (non-complex) systems can be **complicated**.
RECOGNIZE	**Architecture** primarily has to master the **complex** aspects of a system.	**Development** primarily has to master the **complicated** aspects of a system.

“Komplex” bezieht sich auf die extrinsische Schwierigkeit eines Systems auf der höheren Ebene bzw. der Makro-Ebene, da das System aus vielen verschiedenen, miteinander verbundenen Teilen besteht. Es braucht deshalb Zeit, um dieses zu begreifen und als Ganzes zu meistern, obwohl jedes Teil des Systems meist leicht zu erklären ist.

“Kompliziert” bezieht sich auf die intrinsische Schwierigkeit eines Systems auf der unteren Ebene bzw. der Micro-Ebene, da das System viele verschiedene, schwierige Aspekte umfasst. Es braucht deshalb Zeit, um die Teile im Details zu verstehen und zu lernen und jedes Teil des Systems ist meist auch schwer zu erklären.

Beachte: Einfache (nicht-komplizierte) Systeme können komplex sein – offensichtliche (nicht-komplexe) Systeme können kompliziert sein.

Der entscheidende Unterschied ist: Die Architektur bzw. die Konstruktion muss vor allem die komplexen Aspekte eines Systems meistern. Die Entwicklung bzw. die Realisierung muss vor allem die komplizierten Aspekte eines Systems meistern.

Fragen

- ? Muss sich **Architektur** primär um **komplexe** oder **komplizierte** Aspekte eines Systems kümmern?

Architecture Space

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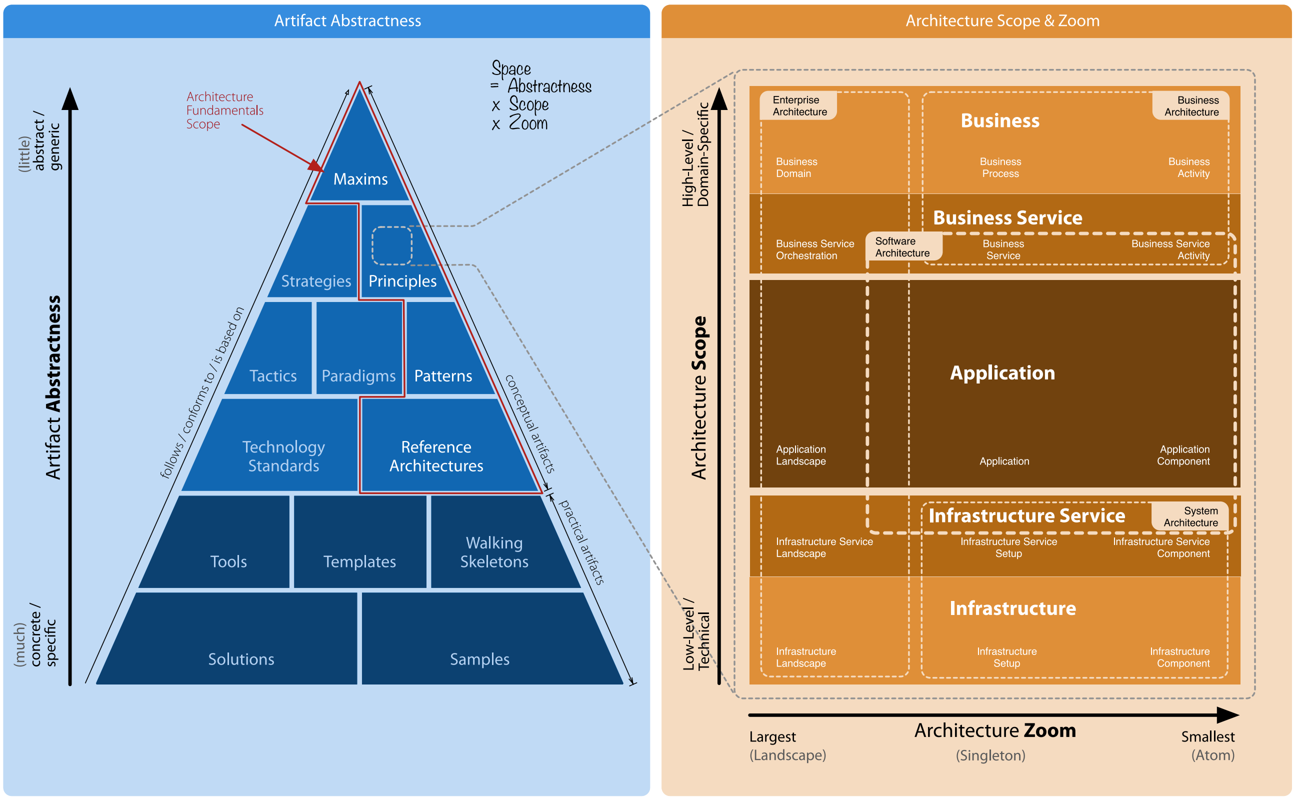

Artifact Abstractness
(little) abstract / generic
Artifact Abstractness
(much) concrete / specific
Architecture Fundamentals Scope
Space = Abstractness x Scope x Zoom
Maxims
Strategies
Principles
Tactics
Paradigms
Patterns
Technology Standards
Reference Architectures
Tools
Templates
Walking Skeletons
Solutions
Samples
follows / conforms to / is based on
conceptual artifacts
practical artifacts
Architecture Scope & Zoom
High-Level / Domain-Specific
Architecture Scope
Low-Level / Technical
Enterprise Architecture
Business
Business Architecture
Business Domain
Business Process
Business Activity
Business Service
Business Service Orchestration
Software Architecture
Business Service
Business Service Activity
Application
Application Landscape
Application
Application Component
Infrastructure Service
System Architecture
Infrastructure Service Landscape
Infrastructure Service Setup
Infrastructure Service Component
Infrastructure
Infrastructure Landscape
Infrastructure Setup
Infrastructure Component
Largest (Landscape)
Architecture Zoom
(Singleton)
Smallest (Atom)

Der **IT Architecture Space** besteht aus drei Dimensionen: der **Artifact Abstractness** (einem Abstraktheitsgrad aller Artefakte, die der Architekt kennt), dem **Architecture Scope** (dem Bereich von Architektur, in dem man agiert) und dem **Architecture Zoom** (der Detailstufe von Architektur, in der man agiert).

Man unterscheidet primär drei Architecture Scopes: (high-level/domain-specific) **Business**, **Application** und (low-level/technical) **Infrastructure**. Zusätzlich werden die zwei sekundären Architecture Scopes **Business Service** und **Infrastructure Service** genutzt, um den "gedanklichen Sprung" von Business zu Application und von Application zu Infrastructure in der Praxis deutlich kleiner und sinnvoller ausfallen zu lassen.

Auf jedem Architecture Scope kann man mindestens drei verschiedene **Architecture Zooms** unterscheiden: von **Landscape** (largest), über **Singleton** bis zu **Atom** (smallest).

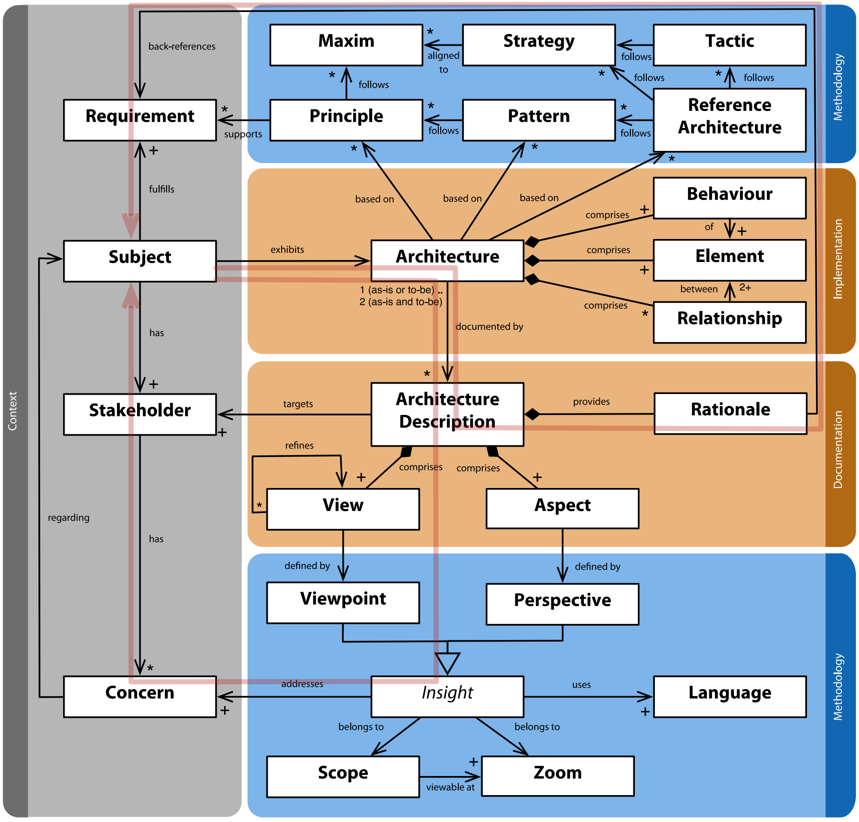
Im Raum aus Architecture Scope und Zoom unterscheidet man vier Arten von **IT Architecture**: **Business Architecture** (Facharchitektur), **Software Architecture** (Software-Architektur), **System Architecture** (System-Architektur) und **Enterprise Architecture** (Unternehmens-Architektur).

Umso höher/niedriger der Abstraktheitsgrad der Artefakte, umso größer/kleiner ist üblicherweise die Abdeckung des Raums aus Architecture Scope und Architecture Zoom.

Fragen

- ? Aus welchen drei Dimensionen besteht der IT Architecture Space?
- ? Welche vier Arten von Architektur kennt man in der IT?

Architecture Ontology


Context
Methodology
Implementation
Documentation
Methodology
Requirement
Maxim
Strategy
Tactic
Principle
Pattern
Reference Architecture
Subject
Architecture
Behaviour
Element
Relationship
Stakeholder
Architecture Description
Rationale
View
Aspect
Viewpoint
Perspective
Concern
Insight
Language
Scope
Zoom
back-references
supports
fulfills
has
regarding
has
aligned to
follows
follows
follows
follows
follows
follows
based on
based on
based on
exhibits
comprises
comprises
comprises
of
between
2+
1 (as-is or to-be) ..
2 (as-is and to-be)
documented by
targets
provides
refines
comprises
comprises
defined by
defined by
addresses
uses
belongs to
belongs to
viewable at
*
+

Requirement: A functional or non-functional demand or imposed obligation on the **Subject**.

Subject: Any type of business process, business service, software application, infrastructure service or infrastructure setup.

Stakeholder: Person, group or entity with an interest in or **Concerns** about the **Architecture**.

Concern: Requirement, objective, intention, or aspiration a **Stakeholder** has on an **Architecture**.

Architecture Description: Set of artifacts that document an **Architecture** in a way **Stakeholders** can understand and ensures their **Concerns** are met.

Viewpoint: Collection of templates and conventions for constructing one type of **View**. It addresses **Concerns** and contains guidelines for constructing a View.

Perspective: Collection of decisions, guidelines and rules that ensure a **Subject** exhibits a set of non-functional **Requirements**, considered across a number of **Views**.

Insight: Superordinate abstract concept to address particular **Concerns** through **Viewpoints** and **Perspectives**.

Architecture: Inherent static and dynamic structure of a **Subject** which comprise **Elements**, the visible **Behaviour** of **Elements** and **Relationships** between **Elements**.

Element: Fundamental part from which a **Subject** can be considered to be constructed.

Behaviour: Run-time characteristic of **Architecture Elements** of the **Subject**.

Relationship: Static or dynamic relationship between **Elements** of an **Architecture** of the **Subject**.

Rationale: Fundamental reasons for a particular chosen **Architecture**, usually strongly based on non-functional **Requirements**.

View: Representation of one or more structural aspects of an **Architecture** that illustrates how the **Architecture** addresses one or more **Concerns**.

Aspect: Representation of one or more non-functional aspects of an **Architecture** that illustrate how the **Architecture** addresses one or more **Concerns**.

Language: Formal, semi-formal or even prose language to describe an Insight, so it optimally addresses the **Concerns** of **Stakeholders**.

Maxim: Fundamental, generally valid set of values and rules to guide the architecture discipline (think: law).

Strategy: Long-term situation-independent plan of approaches to achieve a particular goal (think: war). Aligned and not in conflict with any **Maxims**.

Tactic: Short-term situation-dependent plan of actions to achieve a particular goal (think: battle). Following and supporting a **Strategy**.

Principle: Fundamental truth, rule, tenet or policy an **Architecture** follows.

Pattern: Proven recurring theme, structure, approach or behavior an **Architecture** and its **Elements** can follow.

Reference Architecture: Reusable proven Architecture template, **based** on a set of Patterns and following one or more **Tactics** and **Strategies**.

Scope: Primary area of the **Architecture** space an **Insight** addresses: either Business, Business Service, Application, Infrastructure Service or Infrastructure.

Zoom: Level of view distance to the **Architecture** an **Insight** has. Also known as the **Insight** detail level, ranging from smallest atoms, over singletons to the largest landscape.

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Damit Architekten sinnvoll in der Praxis kommunizieren können, muss man sich auf ein paar wenige Grundbegriffe und deren Bedeutung einigen. Die Begriffe werden in einer Taxonomie definiert und in der **Architecture Ontology** in Beziehung zueinander gesetzt.

In der Architecture Ontology gibt es vor allem zwei wichtige “Schleifen”. Beide starten beim **Subject**, welches eine **Architecture** hat, welche über die **Architecture Description** (im Deutschen “IT-Konzept”) beschrieben wird.

Schleife 1: Die Architecture Description liefert **Rationales** (Begründungen) für Entscheidungen, welche im Idealfall auf **Requirements** zurück referenzieren sind. Denn eine Architecture Description soll nicht das **WAS** sondern vor allem das **WARUM** dokumentieren. Denn das WAS sieht man auch im Code, aber das WARUM nicht!

Schleife 2: Die Architecture Description besteht aus **Views** (Sichten) und **Aspects** (Aspekten), welche methodisch **Viewpoints** und **Perspectives** genannt werden. Beide zusammen liefern **Insights** (Einblicke) auf **Scope** und **Zoom Level** (siehe Architecture Space!) und werden über eine bestimmte (graphische oder textuelle) **Language** dokumentiert. In jedem Fall werden nur solche Insights gegeben, welche einen **Concern** (Anliegen) eines **Stakeholders** (Interessenvertreter) adressieren. Denn man programmiert auch nichts, was nicht gebraucht wird!

Fragen

- Was sollte eine **Architecture Description** (IT-Konzept) neben dem **WAS** vor allem dokumentieren?

- Was sollte eine **Architecture Description** (IT-Konzept) über **Insights** (Einblicke) vor allem addressieren?

ARCHITECTURE FUNDAMENTALS

Requirements Basics

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AF 03.1

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Requirements Specification

A binding document that specifies the requirements for a solution, by focusing on the WHAT and WHY of the solution — and *not* giving instructions for the HOW.

The documented set of requirements has to be: correct, unambiguous, complete, consistent, ranked, verifiable, modifiable, and traceable.

Requirement Classes

FR **Functional** (Shall Do)

A condition or capability that a solution must have to provide its service in terms of its behaviour and information. Think: Functionality.

NFR **Non-Functional** (Shall Be)

A condition, property or quality that a solution must have to satisfy a contract, standard, or other formally imposed obligation. Think: Constraints and "*-ilities".

Requirement Interdependencies

POS **Positive** (Backing)

One requirement supports the other (e.g. for NFRs: Maintainability and Comprehensibility usually support Adaptability, Portability, Modifiability, etc., and Scalability usually supports Availability, etc.)

NEG **Negative** (Trade-Off)

One requirement interferes with the other (e.g. for NFRs: Security usually interferes with Efficiency, Usability, Performance, etc., and Orthogonality can interfere with Usability)

Requirement Characteristics

S **Specific**

The requirement is precise, unambiguous, and clear on what should be done.

M **Measurable**

The requirement can be verified when it has been achieved by use of a particular test.

A **Achievable**

The requirement is achievable given existing circumstances and feasible and viable solutions.

R **Relevant**

The requirement is relevant to the goals of the context.

T **Time-Bound**

The requirement can be achieved within a reasonable time frame.

Requirement Life-Time

E **Enduring**

The requirement lasts forever, as it is derived from core activities and organisational structures.

V **Volatile**

The requirement can be temporary, as it might change over time.

Requirement Expression

[<req-id>] <req-name>:
<subject/actor>
SHALL
<result/action/condition>
BECAUSE
<rationale>

Die **Requirements Specification** (auch Anforderungsspezifikation, Fachkonzept, oder Pflichtenheft) ist ein bindendes Dokument, in dem primär das WAS und das WARUM der Lösung spezifiziert wird, allerdings nicht das konkrete technische WIE. Die Menge an Anforderungen müssen korrekt, eindeutig, vollständig, konsistent, priorisiert, nachprüfbar, änderbar und nachvollziehbar sein.

Es gibt zwei Arten von Requirements: **Functional Requirement** ("Shall Do", Funktionalität) und **Non-Functional Requirements** ("Shall Be", Bedingungen, im Englischen oft über die Wörter mit dem Ende "-ility" ausgedrückt). Der Architekt kümmert sich primär um letztere.

Requirements können außerdem sich gegenseitig positiv (Backing) oder negativ (Trade-Off) beeinflussen. Der Architekt kümmert sich auch hier primär um letztere.

Requirements sollten "SMART" sein: **Specific**, **Measureable**, **Achieveable**, **Relevant** und **Time-Bound**.

Außerdem sind Requirements entweder **Enduring** (feststehend) oder **Volatile** (unbeständig). Der Architekt sollte auf letztere achten.

Fragen

? Welche Art von **Requirements** sollte der Architekt primär im Auge haben und sollten von ihm explizit in der Lösungsfindung addressiert werden?

ARCHITECTURE
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Non-Functional Requirements

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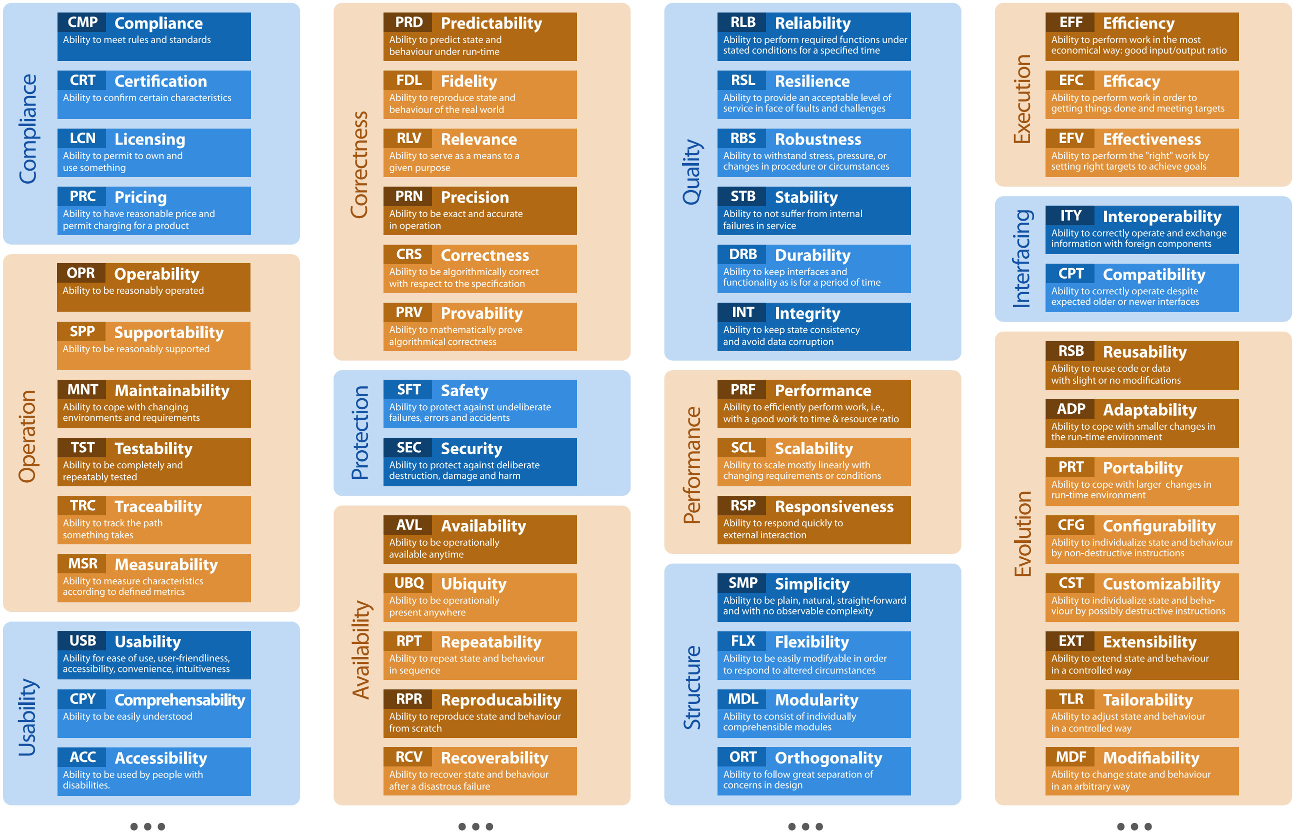

Compliance

CMP **Compliance**
Ability to meet rules and standards

CRT **Certification**
Ability to confirm certain characteristics

LCN **Licensing**
Ability to permit to own and use something

PRC **Pricing**
Ability to have reasonable price and permit charging for a product

Operation

OPR **Operability**
Ability to be reasonably operated

SPP **Supportability**
Ability to be reasonably supported

MNT **Maintainability**
Ability to cope with changing environments and requirements

TST **Testability**
Ability to be completely and repeatably tested

TRC **Traceability**
Ability to track the path something takes

MSR **Measurability**
Ability to measure characteristics according to defined metrics

Usability

USB **Usability**
Ability for ease of use, user-friendliness, accessibility, convenience, intuitiveness

CPY **Comprehensability**
Ability to be easily understood

ACC **Accessibility**
Ability to be used by people with disabilities.

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Correctness

PRD **Predictability**
Ability to predict state and behaviour under run-time

FDL **Fidelity**
Ability to reproduce state and behaviour of the real world

RLV **Relevance**
Ability to serve as a means to a given purpose

PRN **Precision**
Ability to be exact and accurate in operation

CRS **Correctness**
Ability to be algorithmically correct with respect to the specification

PRV **Provability**
Ability to mathematically prove algorithmical correctness

Protection

SFT **Safety**
Ability to protect against undeliberate failures, errors and accidents

SEC **Security**
Ability to protect against deliberate destruction, damage and harm

Availability

AVL **Availability**
Ability to be operationally available anytime

UBQ **Ubiquity**
Ability to be operationally present anywhere

RPT **Repeatability**
Ability to repeat state and behaviour in sequence

RPR **Reproducability**
Ability to reproduce state and behaviour from scratch

RCV **Recoverability**
Ability to recover state and behaviour after a disastrous failure

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Quality

RLB **Reliability**
Ability to perform required functions under stated conditions for a specified time

RSL **Resilience**
Ability to provide an acceptable level of service in face of faults and challenges

RBS **Robustness**
Ability to withstand stress, pressure, or changes in procedure or circumstances

STB **Stability**
Ability to not suffer from internal failures in service

DRB **Durability**
Ability to keep interfaces and functionality as is for a period of time

INT **Integrity**
Ability to keep state consistency and avoid data corruption

Performance

PRF **Performance**
Ability to efficiently perform work, i.e., with a good work to time & resource ratio

SCL **Scalability**
Ability to scale mostly linearly with changing requirements or conditions

RSP **Responsiveness**
Ability to respond quickly to external interaction

Structure

SMP **Simplicity**
Ability to be plain, natural, straight-forward and with no observable complexity

FLX **Flexibility**
Ability to be easily modifyable in order to respond to altered circumstances

MDL **Modularity**
Ability to consist of individually comprehensible modules

ORT **Orthogonality**
Ability to follow great separation of concerns in design

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Execution

EFF **Efficiency**
Ability to perform work in the most economical way: good input/output ratio

EFC **Efficacy**
Ability to perform work in order to getting things done and meeting targets

EFV **Effectiveness**
Ability to perform the "right" work by setting right targets to achieve goals

Interfacing

ITY **Interoperability**
Ability to correctly operate and exchange information with foreign components

CPT **Compatibility**
Ability to correctly operate despite expected older or newer interfaces

Evolution

RSB **Reusability**
Ability to reuse code or data with slight or no modifications

ADP **Adaptability**
Ability to cope with smaller changes in the run-time environment

PRT **Portability**
Ability to cope with larger changes in run-time environment

CFG **Configurability**
Ability to individualize state and behaviour by non-destructive instructions

CST **Customizability**
Ability to individualize state and behaviour by possibly destructive instructions

EXT **Extensibility**
Ability to extend state and behaviour in a controlled way

TLR **Tailorability**
Ability to adjust state and behaviour in a controlled way

MDF **Modifiability**
Ability to change state and behaviour in an arbitrary way

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Non-Functional Requirements gibt es potenziell sehr viele. Für jede Lösung muss deshalb erst die tatsächliche Menge an solchen Requirements festgestellt werden. Diese Menge ist unbedingt vom Architekten zu minimieren!

Bei jedem vertraglich festgeschriebenen Requirements sollte darauf geachtet werden, daß es klar definiert ist, da es große Ähnlichkeiten zwischen Requirements gibt und die Unterschiede teilweise sehr subtil sind.

Ein paar der in der Praxis fast immer zu berücksichtigenden Non-Functional Requirements sind **Maintainability**, **Usability**, **Security**, **Availability**, **Reliability**, **Performance**, **Responsiveness** und **Adaptability**.

Fragen

- Nennen Sie 3 in der Praxis häufig zu berücksichtigende **Non-Functional Requirements**!